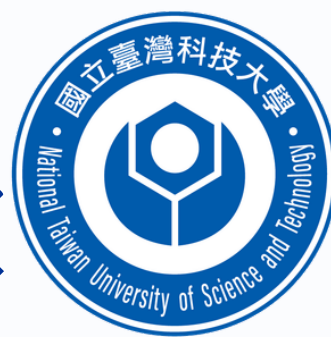




National Taiwan University of Science and Technology

CROWD-SHIPPING PROBLEM WITH TIME WINDOWS, TRANSSHIPMENT NODES, AND DELIVERY OPTIONS

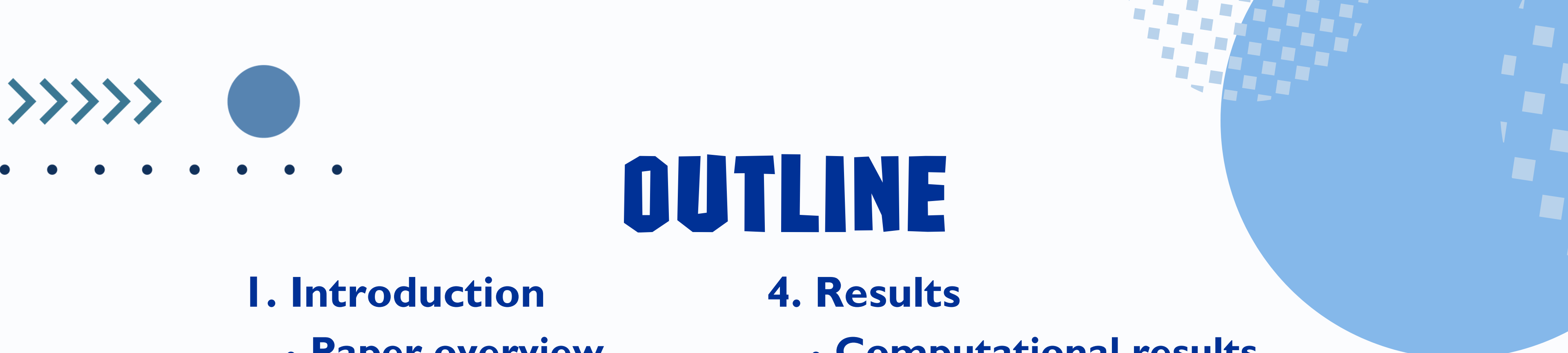
Optimization Models and Applications

Professor: Chun-Chih Chiu

Group Members:

MI1451802 DONG BOI THI

MI1451804 TO NGUYEN TAN PHUONG



OUTLINE

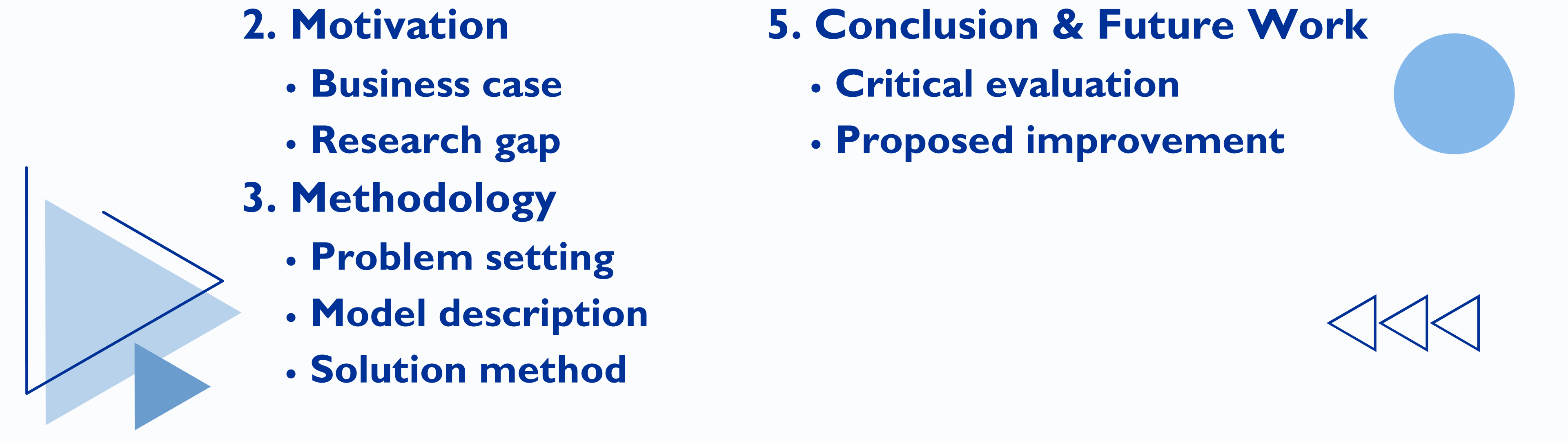
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- Research gap

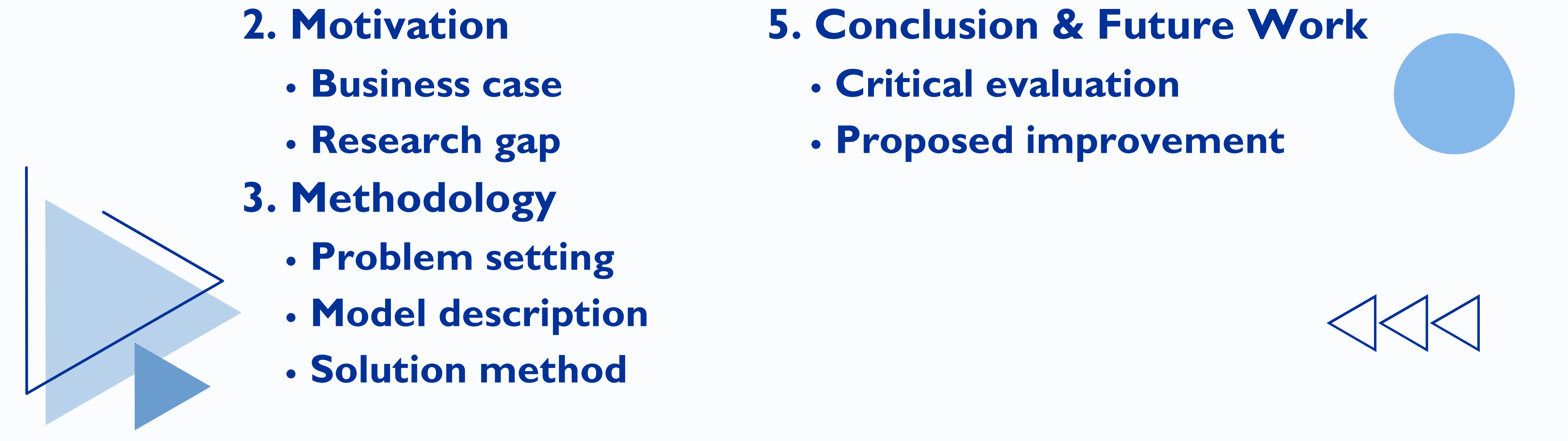
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PAPER OVERVIEW



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Crowd-shipping problem with time windows, transshipment nodes, and delivery options

Vincent F. Yu^{a,b}, Panca Jodiawan^{a,*}, A.A.N. Perwira Redi^c

^a Department of Industrial Management, National Taiwan University of Science and Technology, Taipei 10607, Taiwan
^b Center for Cyber-Physical System Innovation, National Taiwan University of Science and Technology, Taipei 10607, Taiwan
^c Industrial Engineering Department, BINUS Graduate Program – Master of Industrial Engineering, Bina Nusantara University, Jakarta 11480, Indonesia



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ABSTRACT

This research introduces a new variant of the vehicle routing problem in the last-mile delivery process - namely, the Crowd-Shipping Problem with Time Windows, Transshipment Nodes, and Delivery Options (CSPTW-TN-DO). Two types of fleets (i.e., dedicated vehicles and occasional drivers) are available to serve three types of customers. Type 1 customers require a home delivery. The parcel of type 2 customers must be sent to the selected alternative delivery point (ADP). Type 3 customers have the flexibility to either receive their parcel at home or at the selected ADP. Dedicated vehicles are able to serve all types of customers, whereas occasional drivers only make home deliveries. The objective of CSPTW-TN-DO is to minimize the total distribution cost of employing both fleets. We formulate a Mixed Integer Nonlinear Programming (MINLP) model for the problem and solve the model by the commercial solver CPLEX after applying a linearization process. We also propose an Adaptive Large Neighborhood Search (ALNS) to solve a set of newly generated CSPTW-TN-DO instances. The computational results indicate that the proposed ALNS provides high-quality solutions. In addition, we show that the VRPTW with a primary objective of minimizing the total distribution cost is a special case of CSPTW-TN-DO, and that the proposed ALNS achieves comparative performance to the state-of-the-art algorithms for VRPTW. After analyzing several scenarios, we conclude that simultaneously considering occasional drivers, transshipment nodes, and delivery options offers a great opportunity for a last-mile delivery system to reduce its total distribution cost.

Topic

Last-mile delivery optimization with crowd-shipping, TNs, ADPs, and time windows.

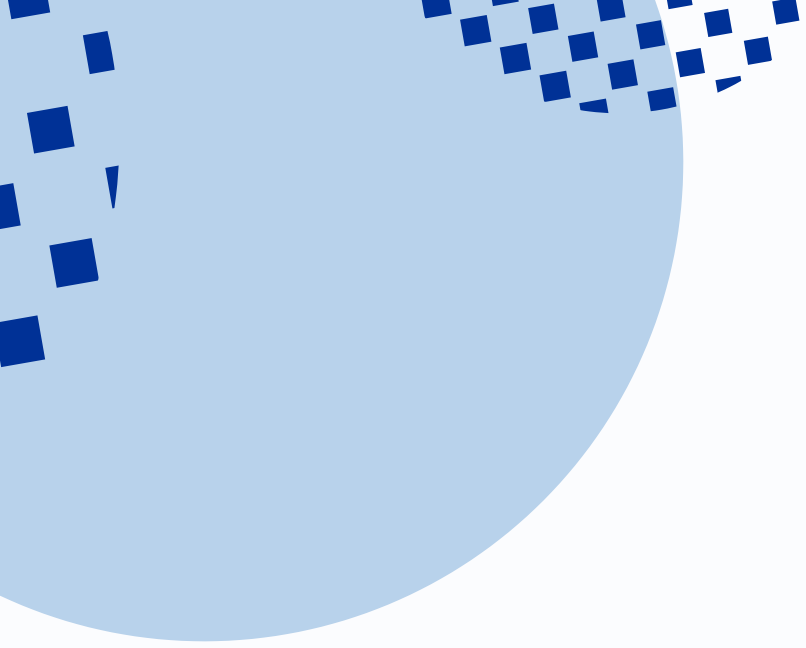
Objective

Minimize total distribution cost.

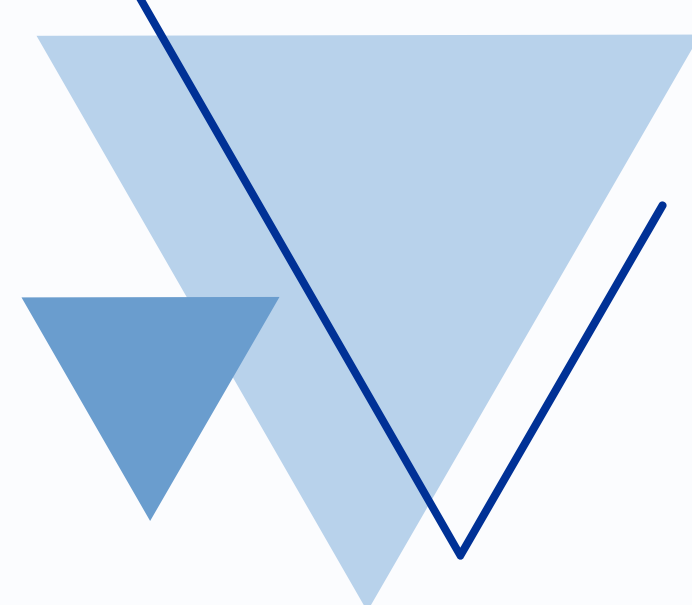
Key decisions

- Where should the parcel be delivered?
- Who should deliver it?
- Which pick-up point and route should be used?

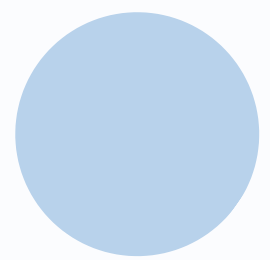




MAIN CONTRIBUTION

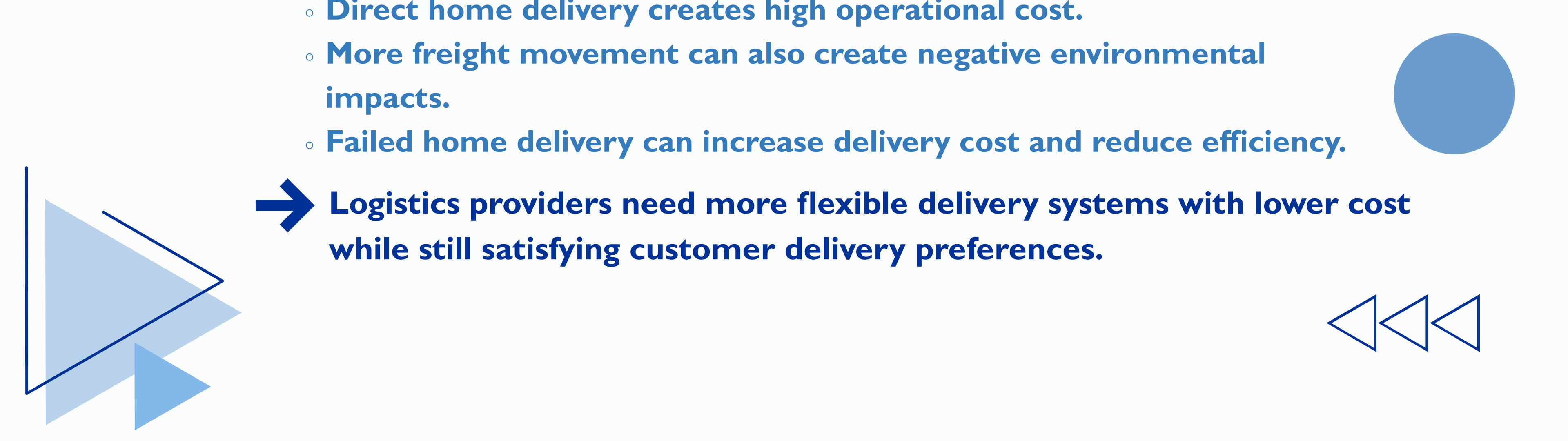


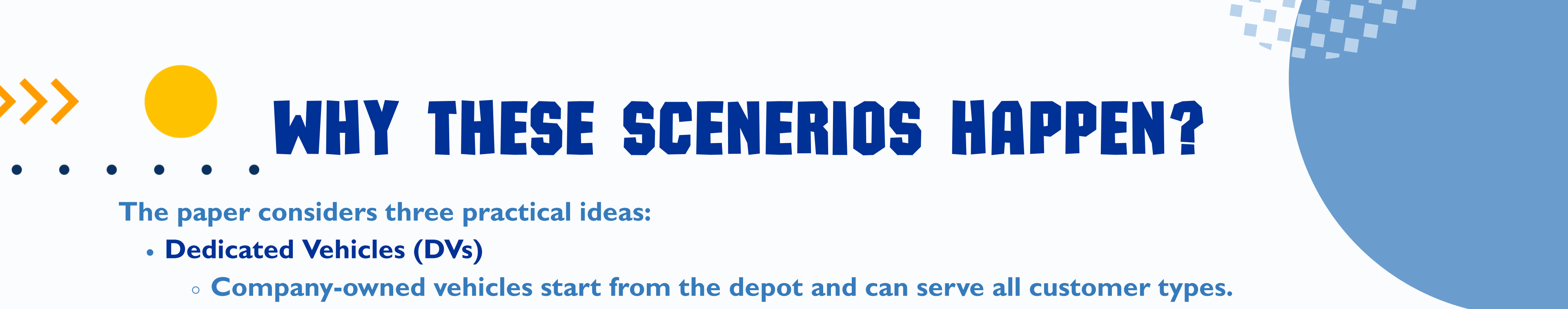
This paper contributes in four main ways:

- 
- Introduces a new Vehicle Routing Problem (**VRP**) variant Crowd-Shipping Problem with Time Windows, Transshipment Nodes, and Delivery Options (**CSPTW-TN-DO**)
 - Formulates the problem as a Mixed Integer Nonlinear Programming (**MINLP**) model
 - Then applies linearization to solve it as a **MILP** using **CPLEX**.
 - Develops an Adaptive Large Neighborhood Search (**ALNS**)
 - Designed specifically for dedicated vehicles, occasional drivers, Alternative delivery points (**ADPs**), and **TNs**.
 - Provides managerial insights
 - Sensitivity analyses are conducted for occasional drivers, dedicated vehicles, transshipment nodes, and delivery options.



BUSINESS PROBLEM

- Last-mile delivery is difficult because:
 - E-commerce growth increases the number of parcels.
 - Customers expect faster delivery.
 - Direct home delivery creates high operational cost.
 - More freight movement can also create negative environmental impacts.
 - Failed home delivery can increase delivery cost and reduce efficiency.
- ➔ **Logistics providers need more flexible delivery systems with lower cost while still satisfying customer delivery preferences.**
- 



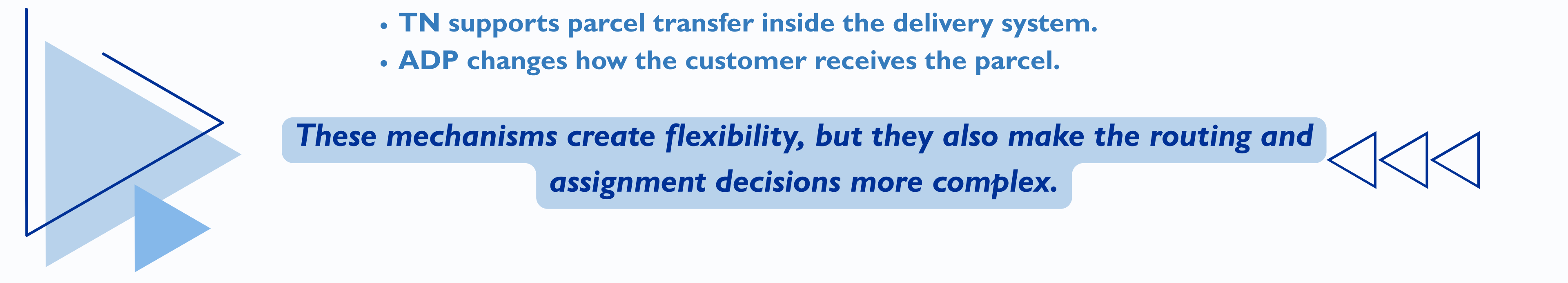
WHY THESE SCENERIOS HAPPEN?

The paper considers three practical ideas:

- **Dedicated Vehicles (DVs)**
 - Company-owned vehicles start from the depot and can serve all customer types.
- **Occasional Drivers (ODs)**
 - Ordinary people, such as commuters or travelers, use extra vehicle capacity to deliver parcels and receive compensation.
- **Alternative Delivery Points (ADPs) and Transshipment Nodes (TNs)**
 - **ADP:** customer pick-up location
 - **TN:** transfer point where occasional drivers can pick up parcels from dedicated vehicles

Key distinction:

- TN supports parcel transfer inside the delivery system.
- ADP changes how the customer receives the parcel.



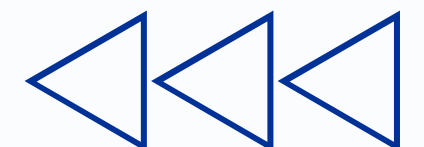
These mechanisms create flexibility, but they also make the routing and assignment decisions more complex.



RESEARCH GAP

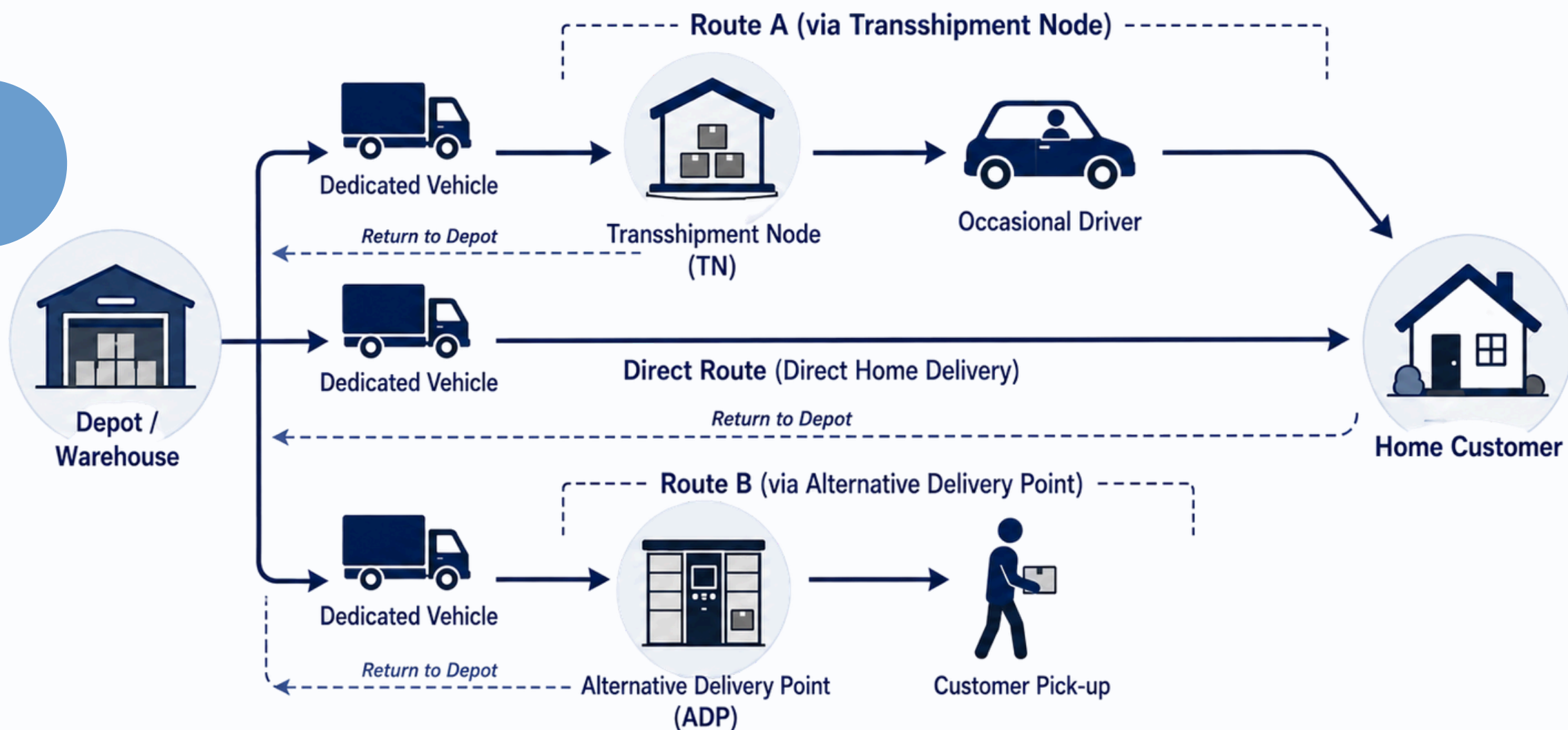
- Previous studies often considered crowd-shipping, transshipment nodes, or alternative delivery points separately.
- **Research Objective:** formulate and solve an integrated last-mile delivery problem that combines crowd-shipping, TNs, ADPs, and time windows to minimize total distribution cost.

Feature	Previous studies	This paper
Occasional drivers	Yes	Yes
Time windows	Some	Yes
Transshipment nodes	Some	Yes
Alternative delivery points	Some	Yes
Integrated model	Limited	Yes



PROBLEM SETTING

Integrated Last-Mile Delivery System



Depot: origin and destination for dedicated vehicles

TNs: transfer points for parcel handoff

ADPs: pick-up points for customers

Customers: demand nodes with different delivery preferences

Occasional drivers (ODs): external drivers with existing trips

Dedicated vehicles (DVs): company-owned vehicles

SERVICE MODE LOGIC

WHO CAN SERVE EACH CUSTOMER TYPE?

Table 1
The available service modes for each type of customers.

Type of Customers	Dedicated Vehicle	Occasional Driver	Alternative Delivery Point
1	●	●	
2			●
3	●	●	●

- **Type 1 (N_h):** Home delivery only.
- **Type 2 (N_a):** ADP delivery only.
- **Type 3 (N_{ha}):** Home delivery or ADP.
- **Highlight:** Type 3 customers can be more flexible so the model can choose the cheaper feasible delivery option cost.



MODEL DESCRIPTION

DECISION LOGIC & VARIABLES

The model determines routing, assignment, and delivery options using a **Mixed Integer Nonlinear Programming (MINLP)** formulation.

Decision question	Key variables	Type	Meaning
Which route should each dedicated vehicle take?	x_{ij}^k	Binary	1 if dedicated vehicle (k) travels from node (i) to node (j)
Which route should each occasional driver take?	y_{ij}^k	Binary	1 if occasional driver (k) travels from node (i) to node (j)
Is a customer served by a dedicated vehicle?	z_i^k	Binary	1 if node/customer (i) is served by dedicated vehicle (k)
Is a customer served by an occasional driver?	r_i^k	Binary	1 if customer (i) is served by occasional driver (k)
Is home delivery selected?	h_i	Binary	1 if customer (i) receives home delivery
Is an ADP selected for a customer?	ε_{ij}^k	Binary	1 if customer (i)'s demand is delivered to ADP (j) by vehicle (k)
What load and time are carried through the route?	$\sigma_{ij}^k, \beta_{ij}^k, \delta_{ik}^D, \delta_i^{CS}$	Continuous	Track vehicle/driver load and arrival time

Key note: Most main decisions are binary yes/no decisions, while load and arrival-time variables are continuous tracking variables.

MODEL DESCRIPTION

OBJECTIVE FUNCTION

Objective: Minimize Total Distribution Cost (Z) to balance company vehicle travel cost and occasional driver compensation cost.

$$\text{Minimize } Z = Z_d + \sum_{k \in CS} Z_{CS}^k$$

- Z_d Total travel distance cost of **Dedicated Vehicles**.
- Z_{CS}^k Compensation cost paid to **Occasional Drivers k** (based on extra distance traveled)



MODEL DESCRIPTION

MAIN CONSTRAINT GROUPS

The full model has many constraints, but they can be grouped into five logical parts.

Constraint group	Main constraints	What they control	Simple meaning
Delivery option constraints	(1)–(4)	Selection of home delivery or ADP delivery	Each customer must be assigned to a valid delivery option based on customer type

Subject to

$$h_i = 1 \quad i \in N_h \quad (1)$$

Type 1 customers must receive home delivery.

$$\sum_{k \in N_{DV}} \sum_{j \in N_L} \varepsilon_{ij}^k = 1 \quad i \in N_a \quad (2)$$

Type 2 customers must be assigned to an ADP.

$$\varepsilon_{ij}^k \leq \gamma_{ij} \quad i \in N_a \cup N_{ha}, k \in N_{DV} \quad (3)$$

Customers can only use their selected ADP.

$$h_i + \sum_{k \in N_{DV}} \sum_{j \in N_L} \varepsilon_{ij}^k = 1 \quad i \in N_{ha} \quad (4)$$

Type 3 customers must choose exactly one option: home delivery or ADP delivery.

MODEL DESCRIPTION

MAIN CONSTRAINT GROUPS

The full model has many constraints, but they can be grouped into five logical parts.

Constraint group	Main constraints	What they control	Simple meaning
Delivery option constraints	(1)–(4)	Selection of home delivery or ADP delivery	Each customer must be assigned to a valid delivery option based on customer type
Vehicle assignment constraints	(5)–(6)	Assignment to dedicated vehicles or occasional drivers	Home-delivery customers are served by either a dedicated vehicle or an occasional driver
Dedicated vehicle routing constraints	(7)–(18)	DV route flow, load, capacity, arrival time, time windows	Dedicated vehicles must follow feasible routes and respect capacity and customer time windows
Occasional driver routing constraints	(19)–(32)	OD pick-up point, route flow, load, capacity, arrival time, time windows	Occasional drivers must pick up parcels at depot/TN and deliver within their own and customer time windows
TN synchronization constraints	(33)–(35)	Handoff between dedicated vehicles and occasional drivers at TNs	An OD can collect parcels at a TN only after a DV has delivered them there

Key difficulty: A route may be short but infeasible if the TN handoff time is not synchronized.

WHY CPLEX ALONE IS NOT ENOUGH

EXACT SOLVER VS PRACTICAL SCALE

The paper formulates the problem as MINLP and then linearizes it into MILP so that CPLEX can solve it.

However:

- The model becomes hard when the number of customers increases.
- CPLEX is useful for small instances as a validation benchmark.
- For larger instances, a metaheuristic is needed.

Small instance results

- 25 customers
- 2 TNs
- 3 ADPs
- 25 occasional drivers
- CPLEX obtains 25 optimal solutions out of 29 small instances.
- ALNS obtains the same objective values but much faster.

Table 6
Computational results for small CSPTW-TN-DO instances.

Instance	CPLEX				ALNS			Gap (Best) %	Gap (Average) %
	UB	LB	Optimality Gap (%)	t (s)	Best Obj	Average Obj	Average CPU time (s)		
C101_25*	147.244	147.244	0.00	3824.88	147.244	147.244	8.85	0.00	0.00
C102_25*	124.883	124.883	0.00	410.47	124.883	124.883	7.19	0.00	0.00
C103_25	146.855	143.805	2.12	10800.00	146.855	146.855	8.92	0.00	0.00
C104_25	142.218	140.452	1.26	10800.00	142.218	143.547	7.64	0.00	0.93
C105_25*	109.75	109.75	0.00	1288.99	109.75	109.750	7.60	0.00	0.00
C106_25*	112.737	112.737	0.00	238.62	112.737	112.737	9.23	0.00	0.00
C107_25*	126.903	126.903	0.00	3093.67	126.903	126.903	6.47	0.00	0.00
C108_25*	138.801	138.801	0.00	1409.95	138.801	144.735	7.27	0.00	4.10
C109_25*	99.255	99.255	0.00	106.96	99.255	99.388	6.17	0.00	0.13
R101_25*	205.127	205.127	0.00	96.67	205.127	205.127	10.15	0.00	0.00
R102_25*	174.482	174.482	0.00	3363.27	174.482	174.482	8.75	0.00	0.00
R103_25*	197.13	197.13	0.00	421.76	197.13	197.130	9.66	0.00	0.00
R104_25*	138.685	138.685	0.00	194.91	138.686	138.686	10.86	0.00	0.00
R105_25	232.711	227.006	2.51	10800.00	232.711	232.874	8.93	0.00	0.07
R106_25*	195.68	195.68	0.00	2896.41	195.68	195.680	7.98	0.00	0.00
R107_25*	197.754	197.754	0.00	2777.19	197.754	197.754	7.61	0.00	0.00
R108_25*	177.305	177.305	0.00	1188.77	177.305	177.305	10.76	0.00	0.00
R109_25*	185.293	185.293	0.00	953.06	185.293	185.293	10.85	0.00	0.00
R110_25*	160.819	160.819	0.00	204.85	160.819	160.819	9.05	0.00	0.00
R111_25*	149.389	149.389	0.00	57.66	149.389	149.389	11.38	0.00	0.00
R112_25*	146.887	146.887	0.00	318.70	146.887	146.887	9.29	0.00	0.00
RC101_25*	221.182	221.182	0.00	609.04	221.182	221.182	10.77	0.00	0.00
RC102_25*	195.706	195.706	0.00	3775.86	195.706	195.706	7.69	0.00	0.00
RC103_25*	242.677	242.677	0.00	7952.96	242.677	242.685	7.98	0.00	0.00
RC104_25*	176.611	176.611	0.00	1317.10	176.611	176.611	9.17	0.00	0.00
RC105_25*	204.51	204.51	0.00	2910.89	204.51	204.510	8.14	0.00	0.00
RC106_25*	166.227	166.227	0.00	889.99	166.227	166.227	6.90	0.00	0.00
RC107_25*	193.282	193.282	0.00	905.18	193.282	193.282	7.80	0.00	0.00
RC108_25*	170.76	170.76	0.00	6047.44	170.76	170.760	7.04	0.00	0.00
Average			0.20	2746.73			8.62	0.00	0.18

PROPOSED METHOD: ADAPTIVE LARGE NEIGHBORHOOD SEARCH (ALNS)

2. Why this is not a generic ALNS?

ALNS solves the problem by repeatedly improving a current delivery plan.

Its core idea is “destroy and repair”, combined with local search, simulated annealing acceptance, and adaptive operator selection.

1. Overall main procedure

1. **Generate an initial solution.**
2. **Select a destroy operator.**
3. **Remove part of the current solution.**
4. **Select a repair operator.**
5. **Reinsert removed customers.**
6. **Apply local search if the solution is promising.**
7. **Accept or reject the new solution using simulated annealing.**
8. **Update operator probabilities based on past performance.**

S_1^1	0	27	19	11	10	7	8	5	17	18	13	0
S_2^1	0	0	29	29	29	29	29	29	29	29	0	0
S_1^2	0	21	6	16	14	23	22	4	1	25	24	12
S_2^2	0	28	28	28	28	0	0	30	30	30	30	0

(a)

C^9	O9	27	9	D9	
C^{11}	O11	0	2	15	D11
C^{17}	O17	0	3	D17	
C^{19}	O19	27	20	D19	

(b)

Fig 2. (a) The solution representation for the utilized dedicated vehicles in Fig. 1; (b) The solution representation for the utilized occasional drivers in Fig. 1.

Customization aspect	Main content
Problem-specific destroy operators	Random ADP removal; Random TN removal; Worst ADP removal; Related removal considering delivery type
Problem-specific repair logic	Best insertion and regret insertion; Special insertion strategies for occasional driver routes; Feasibility checks for time windows and TN synchronization
Local search	Move and swap operators; Works across dedicated vehicle routes and occasional driver routes

Algorithm 2: The overall structure of the ALNS heuristic

Input: $\mathbb{O}_R, \mathbb{O}_I, \eta_{max}, \eta_s, \vartheta_{ls}, T_0, \alpha$, and an instance of CSPTW-TN-DO
Output: S^*

```
1  $S^c \leftarrow InitialSolution$ 
2  $S^* \leftarrow S^c$ 
3  $\pi_r, \pi_i \leftarrow InitializeProbabilities(\pi_r, \pi_i)$ 
4  $T \leftarrow T_0, i \leftarrow 1$ 
5 while  $i \leq \eta_{max}$  do
6   Select a removal and a repair operator based on  $\pi_r$  and  $\pi_i$  where
7    $r \in \mathbb{O}_R$  and  $i \in \mathbb{O}_I$ 
8    $S^w \leftarrow Repair(Destroy(S^c))$ 
9   if  $f(S^w) \leq (1 + \delta_{ls})f(S^*)$  then
10     $S^w \leftarrow LocalSearch(S^w)$ 
11    if  $f(S^w) < f(S^c)$  then
12       $S^c \leftarrow S^w$ 
13      if  $f(S^c) < f(S^*)$  then
14         $S^* \leftarrow S^c$ 
15    else
16       $S^c \leftarrow accept(S^w, S^c)$ 
17    if  $(modulo(i, \eta_s) \equiv 0)$  then
18       $\pi_r, \pi_i \leftarrow UpdateOperatorProbabilities(\pi_r, \pi_i)$ 
19       $T \leftarrow \alpha T$ 
20       $i \leftarrow i + 1$ 
21 return  $S^*$ 
```

Table 10

Contribution of each destroy operator and repair operator for obtaining best solutions.

Removal Operator	Percentage (%)	Repair Operator	Percentage (%)
Random	5.75	Best Insertion	9.37
Random ADP	3.56	Perturbed Best Insertion	7.23
Random TN	3.92	Regret 2 Insertion	20.27
Deterministic worst customer	4.11	Perturbed Regret 2 Insertion	21.21
Worst ADP	4.52	Regret 3 Insertion	20.56
Probabilistic worst customer	13.95	Perturbed Regret 3 Insertion	21.36
Random route	14.36		
Deterministic related	7.16		
Probabilistic related	8.21		
Historical node	12.29		
Neighborhood	9.41		
Node neighborhood	12.75		

Highlight: regret insertion operators contribute more than best insertion.

BENCHMARK SOURCE & INSTANCE DESIGN

FROM SOLOMON VRPTW INSTANCES TO GENERATED BENCHMARK SETS

Since **CSPTW-TN-DO** is a new **VRP** variant, the authors generated new benchmark instances based on Solomon's **VRPTW** benchmark instances plus additional crowd-shipping elements.

Item	Small Instances	Large Instances
Base dataset	Solomon VRPTW C1/R1/RC1	Solomon VRPTW C1/R1/RC1
Number of instances	29	29
Customers	25	100
TNs	2	4
ADPs	3	5
Occasional drivers	25	50
Total	58 generated CSPTW-TN-DO instances	58 generated CSPTW-TN-DO instances

Benchmark Generation Flow



Solomon VRPTW instances
C1 / R1 / RC1



Select small and large instances
25 or 100 customers



Add CSPTW-TN-DO elements
customer types + TNs + ADPs + occasional drivers



Generated CSPTW-TN-DO benchmark
58 instances

BENCHMARK SOURCE & INSTANCE DESIGN

FROM VRPTW TO CSPTW-TN-DO DATA

To transform a standard **VRPTW** instance into a **CSPTW-TN-DO** instance, the authors added several new data components.

Added element	Why it is needed
Customer types	Define home delivery / ADP delivery / flexible delivery
TN locations	Allow parcel transfer from dedicated vehicles to ODs
ADP locations	Allow customer pick-up and parcel consolidation
Customer-ADP assignment	Assign type 2/type 3 customers to closest ADP
OD origins and destinations	Represent real crowd-shipping trips
OD time windows	Define when occasional drivers are available
OD capacity	Limit each OD to at most 2 customers
Service time per unit demand	Link service time with customer demand

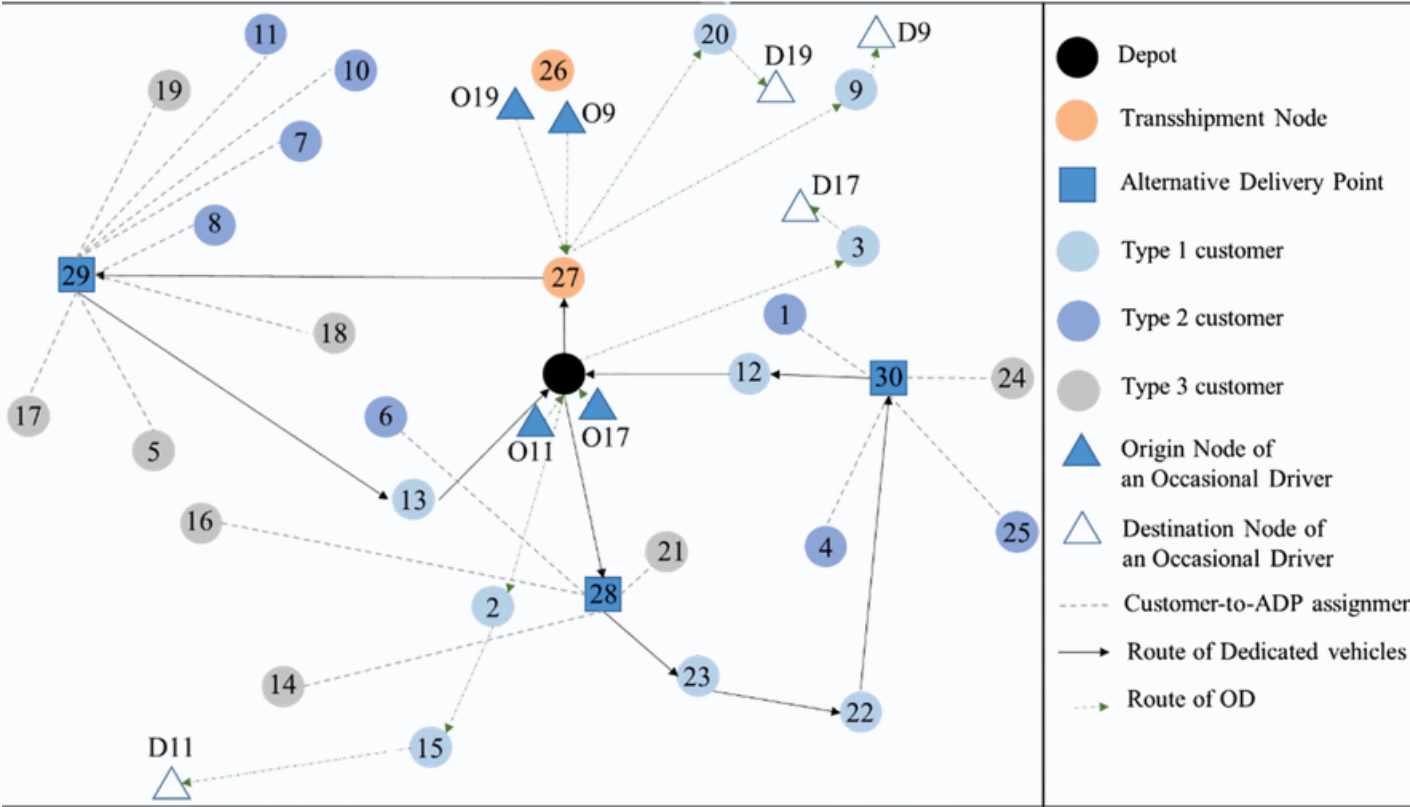


Fig 1. An illustrative example of a CSPTW-TN-DO solution.

Figure 1 shows actual CSPTW-TN-DO solution with depot, TNs, ADPs, customer types, OD origins/destinations

COMPUTATIONAL RESULTS

Test purpose	Dataset / benchmark	Main comparison	Key results	Meaning
1. Validate ALNS on standard VRPTW (Table 3)	Solomon’s 100-customer VRPTW benchmark	ALNS vs best-known / optimal solutions	ALNS obtains 32/56 best-known or optimal solutions; average gap = 0.17%	ALNS is reliable on a standard routing benchmark before being used for the new problem
2. Compare ALNS with exact solver (Table 6)	Small CSPTW-TN-DO instances: 25 customers, 2 TNs, 3 ADPs, 25 ODs	ALNS vs CPLEX	CPLEX obtains 25/29 optimal solutions; ALNS obtains the same objective values and solves each small instance within 12 seconds	CPLEX validates quality, but ALNS is much faster
3. Test ALNS on larger CSPTW-TN-DO (Table8)	Large CSPTW-TN-DO instances: 100 customers, 4 TNs, 5 ADPs, 50 ODs	ALNS performance over 10 runs	Average CPU time = 2.61 min ; average gap between average and best solution = 2.29% ; max gap ≤ 4.89%	ALNS provides robust solutions in reasonable time for larger cases

The authors first validate **ALNS** on a standard **VRPTW** benchmark, then use **CPLEX** as a quality benchmark for small **CSPTW-TN-DO** instances, and finally apply **ALNS** to larger instances where exact solving becomes less practical.

Table 3 First comparison results on Solomon’s VRPTW instances with 100 customer nodes.					
Instance	ALNS		BKS ¹	Ref	Gap (%)
	Best Obj	Average CPU Time (s)			
C101	827.3	56.86	827.3*	M	0.00
C102	827.3	57.62	827.3*	M	0.00
C103	826.3	55.89	826.3*	M	0.00
C104	822.9	54.71	822.9*	M	0.00
C105	827.3	59.30	827.3*	M	0.00
C106	827.3	57.84	827.3*	M	0.00
C107	827.3	56.67	827.3*	M	0.00
C108	827.3	56.73	827.3*	M	0.00
C109	827.3	57.91	827.3*	M	0.00
R101	1645	66.98	1637.7*	M	0.45
R102	1472.4	68.67	1466.6*	M	0.40
R103	1208.7	68.43	1208.7*	M	0.00
R104	976	57.03	971.5*	M	0.46
R105	1355.3	63.02	1355.3*	M	0.00
R106	1234.6	63.21	1234.6*	M	0.00
R107	1065.6	61.52	1064.6*	M	0.09
R108	939.1	60.90	932.1*	M	0.75
R109	1146.9	61.95	1146.9*	M	0.00
R110	1075.1	58.48	1068*	M	0.66
R111	1049.6	58.67	1048.7*	M	0.09
R112	953.9	56.70	948.6*	M	0.56
RC101	1621.2	51.09	1619.8*	M	0.09
RC102	1457.4	56.94	1457.4*	M	0.00
RC103	1258.2	57.17	1258*	M	0.02
RC104	1132.3	54.72	1132.3*	M	0.00
RC105	1515	57.17	1513.7*	M	0.09
RC106	1373.9	56.21	1372.7*	M	0.09
RC107	1209.3	57.99	1207.8*	M	0.12
RC108	1131.7	54.98	1114.2*	M	1.57
C201	589.1	68.17	589.1*	M	0.00
C202	589.1	65.76	589.1*	M	0.00
C203	588.7	71.11	588.7*	M	0.00
C204	588.1	73.54	588.1*	M	0.00
C205	586.4	68.44	586.4*	M	0.00
C206	586	70.43	586*	M	0.00
C207	585.8	67.97	585.8*	M	0.00
C208	585.8	68.74	585.8*	M	0.00
R201	1143.2	70.32	1143.2*	M	0.00
R202	1036.9	71.35	1029.6*	M	0.71
R203	877.1	69.02	870.8*	M	0.72
R204	732.9	71.35	731.3*	B	0.22
R205	949.8	72.67	949.8*	M	0.00
R206	879.9	73.56	875.9*	M	0.46
R207	794	73.20	794*	M	0.00
R208	701	72.17	701.2	D	0.00
R209	855.7	69.26	854.8*	M	0.11
R210	906.1	76.14	900.5*	M	0.62
R211	746.7	68.01	746.7*	B	0.00
RC201	1265.9	68.42	1261.8*	M	0.32
RC202	1097.1	66.47	1092.3*	M	0.44
RC203	923.7	67.36	923.7*	M	0.00
RC204	783.5	70.75	783.5*	B	0.00
RC205	1154	70.30	1154*	M	0.00
RC206	1051.1	67.56	1051.1*	M	0.00
RC207	966.5	66.22	962.9*	M	0.37
RC208	777.9	63.94	776.1*	B	0.23
Average					0.17
Bold value indicates that we obtain the BKS solution.					

Table 6 Computational results for small CSPTW-TN-DO instances.									
Instance	CPLEX			t (s)	ALNS		Average CPU time (s)	Gap (Best) %	Gap (Average) %
	UB	LB	Optimality Gap (%)		Best Obj	Average Obj			
C101_25*	147.244	147.244	0.00	3824.88	147.244	147.244	8.85	0.00	0.00
C102_25*	124.883	124.883	0.00	410.47	124.883	124.883	7.19	0.00	0.00
C103_25	146.855	143.805	2.12	10800.00	146.855	146.855	8.92	0.00	0.00
C104_25	142.218	140.452	1.26	10800.00	142.218	143.547	7.64	0.00	0.93
C105_25*	109.75	109.75	0.00	1288.99	109.75	109.750	7.60	0.00	0.00
C106_25*	112.737	112.737	0.00	238.62	112.737	112.737	9.23	0.00	0.00
C107_25*	126.903	126.903	0.00	3093.67	126.903	126.903	6.47	0.00	0.00
C108_25*	138.801	138.801	0.00	1409.95	138.801	144.735	7.27	0.00	4.10
C109_25*	99.255	99.255	0.00	106.96	99.255	99.388	6.17	0.00	0.13
R101_25*	205.127	205.127	0.00	96.67	205.127	205.127	10.15	0.00	0.00
R102_25*	174.482	174.482	0.00	3363.27	174.482	174.482	8.75	0.00	0.00
R103_25*	197.13	197.13	0.00	421.76	197.13	197.130	9.66	0.00	0.00
R104_25*	138.685	138.685	0.00	194.91	138.686	138.686	10.86	0.00	0.00
R105_25	232.711	227.006	2.51	10800.00	232.711	232.874	8.93	0.00	0.07
R106_25*	195.68	195.68	0.00	2896.41	195.68	195.680	7.98	0.00	0.00
R107_25*	197.754	197.754	0.00	2777.19	197.754	197.754	7.61	0.00	0.00
R108_25*	177.305	177.305	0.00	1188.77	177.305	177.305	10.76	0.00	0.00
R109_25*	185.293	185.293	0.00	953.06	185.293	185.293	10.85	0.00	0.00
R110_25*	160.819	160.819	0.00	204.85	160.819	160.819	9.05	0.00	0.00
R111_25*	149.389	149.389	0.00	57.66	149.389	149.389	11.38	0.00	0.00
R112_25*	146.887	146.887	0.00	318.70	146.887	146.887	9.29	0.00	0.00
RC101_25*	221.182	221.182	0.00	609.04	221.182	221.182	10.77	0.00	0.00
RC102_25*	195.706	195.706	0.00	3775.86	195.706	195.706	7.69	0.00	0.00
RC103_25*	242.677	242.677	0.00	7952.96	242.677	242.685	7.98	0.00	0.00
RC104_25*	176.611	176.611	0.00	1317.10	176.611	176.611	9.17	0.00	0.00
RC105_25*	204.51	204.51	0.00	2910.89	204.51	204.510	8.14	0.00	0.00
RC106_25*	166.227	166.227	0.00	889.99	166.227	166.227	6.90	0.00	0.00
RC107_25*	193.282	193.282	0.00	905.18	193.282	193.282	7.80	0.00	0.00
RC108_25*	170.76	170.76	0.00	6047.44	170.76	170.760	7.04	0.00	0.00
Average			0.20	2746.73			8.62	0.00	0.18

Table 8 Computational results for large CSPTW-TN-DO instances.				
Instance	Best Obj	AverageObj	Gap ^h (%)	Average CPU time (min)
c101	781.032	803.001	2.74	2.68
c102	769.298	778.959	1.24	3.13
c103	580.802	588.838	1.36	2.74
c104	553.352	558.367	0.90	2.99
c105	665.679	679.711	2.06	2.54
c106	782.135	801.622	2.43	2.58
c107	652.622	686.157	4.89	2.63
c108	565.389	587.558	3.77	2.63
c109	464.665	469.446	1.02	2.26
r101	659.029	682.653	3.46	2.02
r102	588.021	601.901	2.31	2.38
r103	533.571	559.640	4.66	2.91
r104	452.998	458.600	1.22	2.67
r105	636.921	646.510	1.48	2.30
r106	553.385	560.209	1.22	2.16
r107	551.319	558.023	1.20	2.34
r108	557.601	564.453	1.21	3.20
r109	482.675	489.176	1.33	2.25
r110	491.891	513.247	4.16	2.26
r111	586.39	593.169	1.14	2.74
r112	446.254	453.228	1.54	3.70
rc101	784.131	794.733	1.33	2.13
rc102	666.729	689.551	3.31	2.59
rc103	626.712	657.724	4.72	2.82
rc104	562.217	566.124	0.69	3.17
rc105	666.346	690.189	3.45	2.34
rc106	769.152	785.730	2.11	2.30
rc107	633.425	651.480	2.77	2.39
rc108	632.031	648.654	2.56	2.70
Average	610.199	624.781	2.29	2.61

KEY MANAGERIAL INSIGHTS

WHAT CAN MANAGERS LEARN FROM THE MODEL?

Key insight	Evidence from paper	Managerial meaning
1. ADPs and TNs reduce distribution cost (Fig. 3a)	ADP reduces cost by 25.41% with TNs and 28.88% without TNs. TNs reduce cost by 7.21% without ADPs and 2.68% with ADPs.	Cost saving comes from combining customer pick-up flexibility and better use of occasional drivers.
2. More ADPs are not always better (Fig. 5a)	When ADPs increase from 3 to 7, total distribution cost increases from 7572.334 to 8271.478 in the tested setting.	More pick-up points may reduce consolidation efficiency and increase dedicated vehicle travel cost.
3. Customer flexibility matters (Fig,. 5b)	The highest cost occurs when Type 1 customers are dominant: 8394.409. The lowest cost occurs when Type 3 customers are dominant: 7287.976.	Flexible customers give the system more routing and delivery options, which helps reduce cost.

The best cost reduction does not come from one single feature. It comes from coordinating ADPs, TNs, occasional drivers, and flexible customer delivery options.

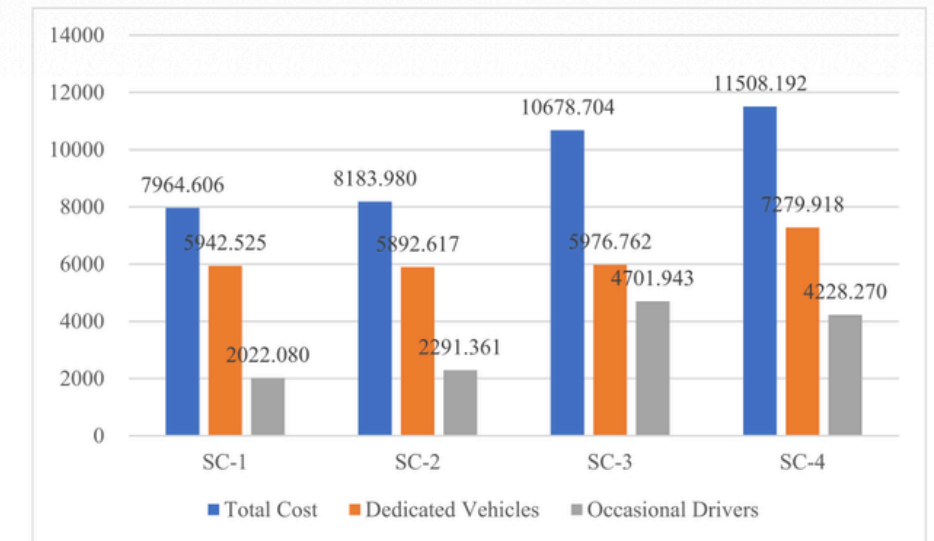


Fig. 3. Changes under four different scenarios: (a) Distribution costs, and (b) Number of employed occasional drivers and the number of customers served by occasional drivers

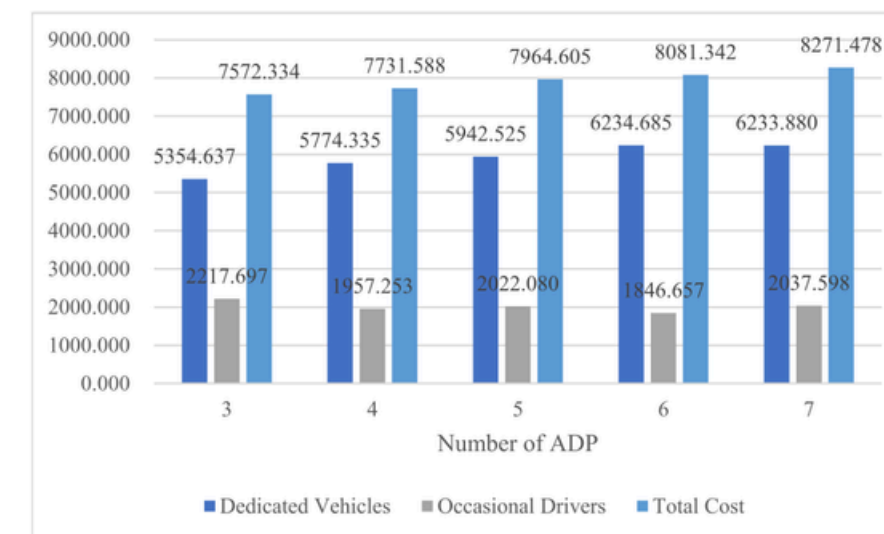


Fig. 5. Changes under varying numbers of ADPs: (a) Distribution costs, and (b) Number of type 1 customers served by occasional drivers

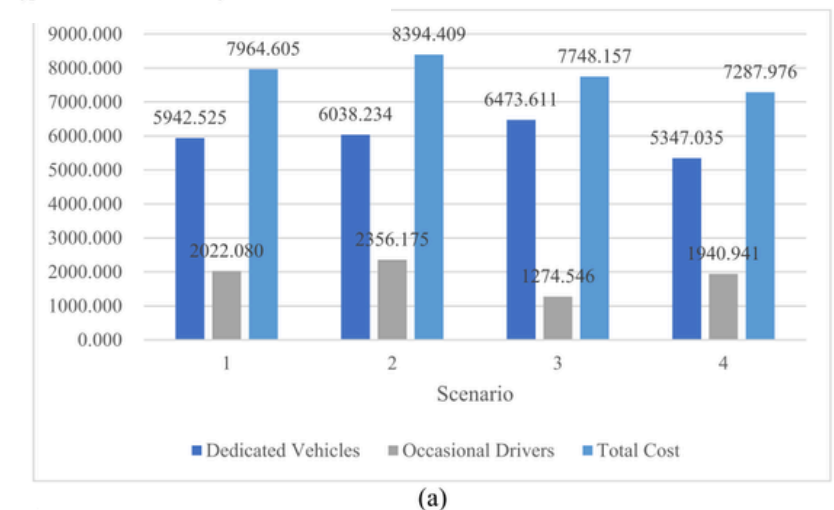


Fig. 6. Changes under different scenarios of types of customers: (a) Distribution Costs, and (b) Number of employed occasional drivers and number of served customers

CRITICAL EVALUATION

Strengths	Limitations
Integrates multiple realistic last-mile delivery mechanisms	Benchmark instances are generated, not real company data
Provides a clear mathematical formulation	Focuses mainly on total distribution cost
Uses CPLEX to validate small instances	Occasional drivers are assumed homogeneous
Develops ALNS for scalable solution	Environmental impact is not included in the objective
Provides useful managerial insights	Time-dependent urban traffic is not considered
	The full model is complex and difficult to replicate exactly

The paper is strong in model design, solution approach, and managerial insight, but it is limited by generated data, simplifying assumptions, and a cost-only objective.

CONCLUSION

This paper provides an integrated optimization model for last-mile delivery.

Main takeaway

- Crowd-shipping can reduce delivery cost by using occasional drivers.
- ADPs reduce home-delivery burden through consolidation.
- TNs improve the usability of occasional drivers.
- ALNS provides high-quality solutions faster than exact solving for larger cases.
- The best system design depends on customer flexibility, compensation factor, ADP placement, and time-window tightness.

Final message

The paper's main value is not only a new routing algorithm. It shows how optimization can support practical last-mile delivery decisions by coordinating company vehicles, crowd-shippers, transshipment nodes, and customer delivery flexibility.

PROPOSED IMPROVEMENT

The original paper minimizes total distribution cost.

$$\min Z = Z_D + \sum_{k \in CS} Z_k^{CS}$$

Proposed extension: **Cost + Emission Objective**

$$\min Z' = Z_D + \sum_{k \in CS} Z_k^{CS} + \lambda \times E$$

Where:

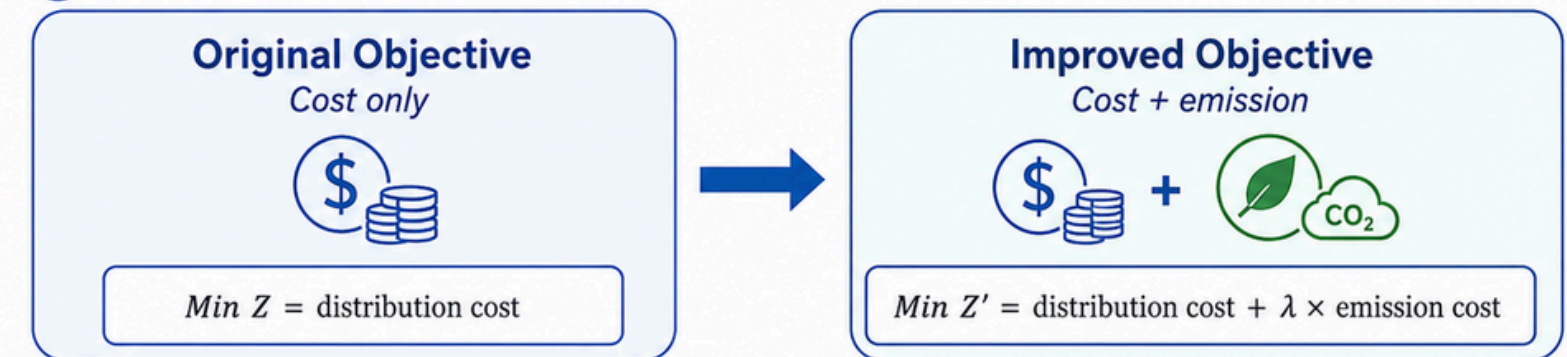
- E : estimated emission cost
- λ : weight for environmental impact

Why this improvement is reasonable

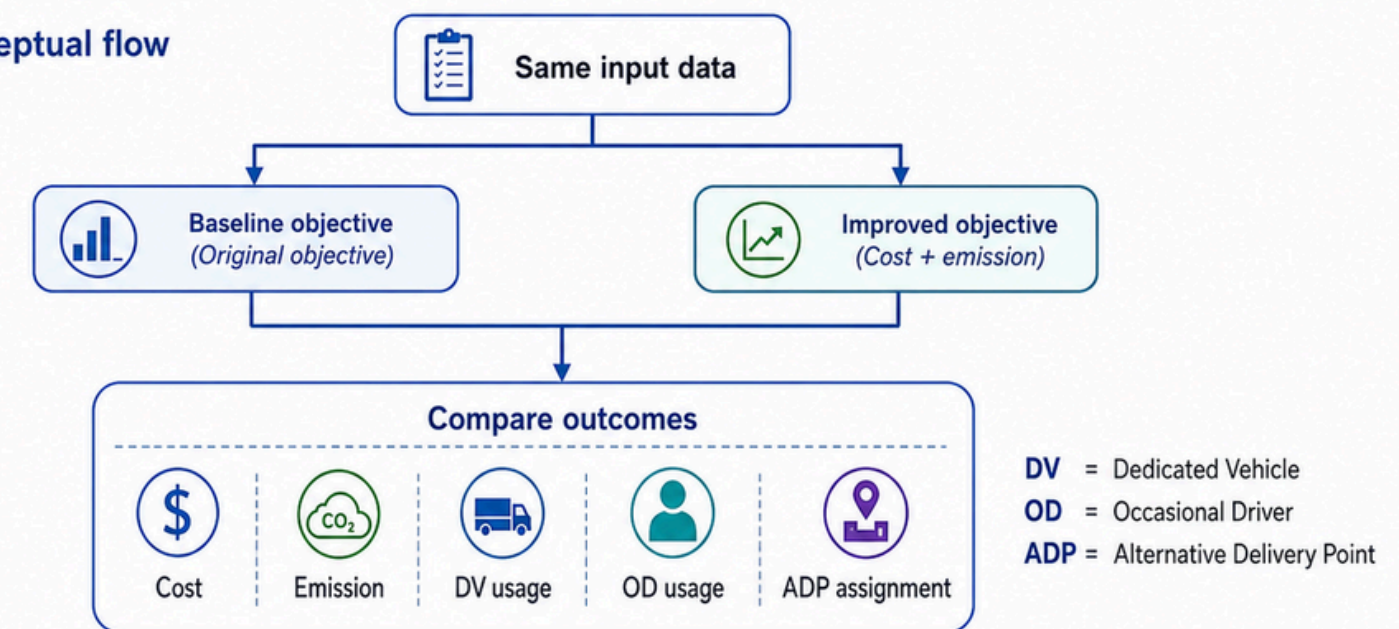
- The motivation section mentions negative environmental impacts from freight movement.
- The conclusion suggests a multi-objective version considering transportation cost and emission cost.
- It keeps the original model structure but adds a sustainability objective.

Proposed Improvement: Objective Comparison

1 Objective comparison

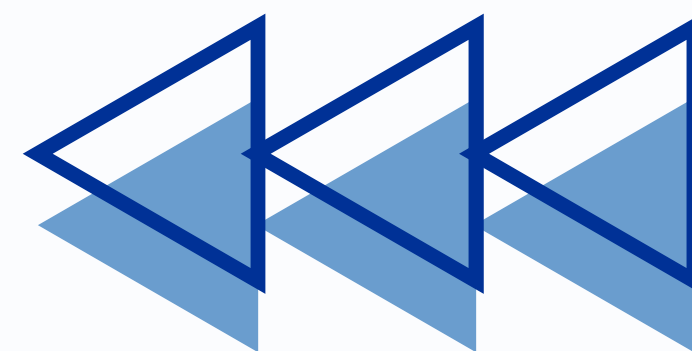
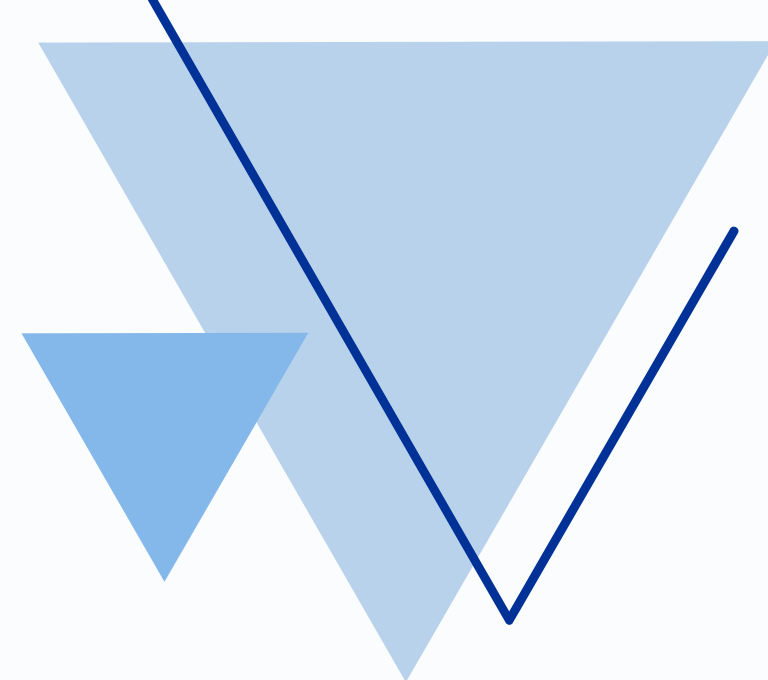


2 Conceptual flow





Q&A





THANK YOU

